

ECE 332: Engineering Electromagnetics II

Lecture 13

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Quick Review

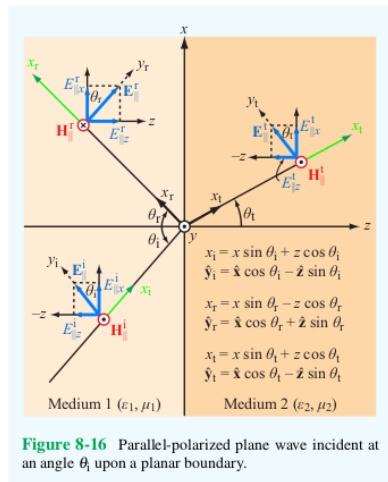
Parallel Polarization

For an incident wave ($\mathbf{E}_{\parallel}^i, \mathbf{H}_{\parallel}^i$) that is **polarized parallel** to the plane of incidence, the phasors of the EM fields of the **incident wave** are

$$\tilde{\mathbf{E}}_{\parallel}^i = E_{\parallel 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \cdot (\hat{\mathbf{x}} \cos \theta_i - \hat{\mathbf{z}} \sin \theta_i), \quad (1)$$

$$\tilde{\mathbf{H}}_{\parallel}^i = \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \hat{\mathbf{y}}. \quad (2)$$

The **tangential boundary conditions** imply Snell's laws and relationships between the amplitudes and angles of the reflected and transmitted waves and those of the incident wave.



Quick Review (Continued)

The phasors of the EM fields of the **reflected wave** are

$$\tilde{\mathbf{E}}_{\parallel}^r = \Gamma_{\parallel} E_{\parallel 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} (\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (3)$$

$$\tilde{\mathbf{H}}_{\parallel}^r = -\Gamma_{\parallel} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \hat{\mathbf{y}}, \quad (4)$$

and the phasors of the EM fields of the **transmitted wave** are

$$\tilde{\mathbf{E}}_{\parallel}^t = \tau_{\parallel} E_{\parallel 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} (\hat{\mathbf{x}} \cos \theta_t - \hat{\mathbf{z}} \sin \theta_t), \quad (5)$$

$$\tilde{\mathbf{H}}_{\parallel}^t = \tau_{\parallel} \frac{E_{\parallel 0}^i}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} \hat{\mathbf{y}}, \quad (6)$$

where the **Fresnel reflection coefficient for parallel polarization** is

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (7)$$

Quick Review (Continued)

the **Fresnel transmission coefficient for parallel polarization** is

$$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (8)$$

and **Snell's law of refraction** implies

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_1}{k_2} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}. \quad (9)$$

Note that the boundary conditions also imply **Snell's law of reflection**

$$\theta_r = \theta_i. \quad (10)$$

In this case,

$$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}. \quad (11)$$

Quick Review (Continued)

Special cases:

- When **Medium 2 is a perfect conductor**, $\eta_2 = 0$ and $\Gamma_{\parallel} = -1$ and $\tau_{\parallel} = 0$.
- When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$ and

$$\Gamma_{\parallel} = \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}. \quad (12)$$

Quick Review (Continued)

Brewster Angle

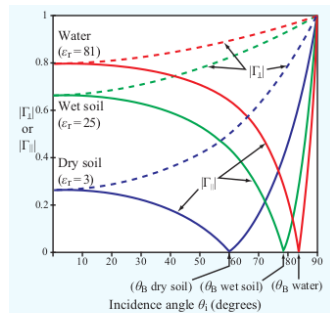
There is an angle of incidence $\theta_{B\parallel}$ for EM waves **polarized parallel to the plane of incidence** where $\Gamma_{\parallel} = 0$. This angle is called the **Brewster angle**. It satisfies

$$\sin \theta_{B\parallel} = \sqrt{\frac{1 - (\epsilon_1 \mu_2 / \epsilon_2 \mu_1)}{1 - (\epsilon_1 / \epsilon_2)^2}}. \quad (13)$$

When $\mu_1 = \mu_2$,

$$\theta_{B\parallel} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}. \quad (14)$$

When $\theta_i = \theta_{B\parallel}$, the parallel polarized component is **completely transmitted** into Medium 2 and **the reflected wave is perpendicularly polarized**, thereby “polarizing” the wave. So, the Brewster angle is also called the **polarizing angle**.



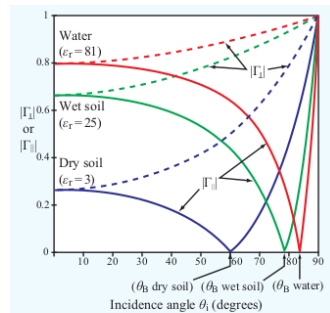
Quick Review (Continued)

When $\mu_1 \neq \mu_2$, there is also a Brewster angle $\theta_{B\perp}$ for EM waves **polarized perpendicular to the plane of incidence** where $\Gamma_{\perp} = 0$. It is given by

$$\sin \theta_{B\perp} = \sqrt{\frac{1 - (\mu_1 \epsilon_2 / \mu_2 \epsilon_1)}{1 - (\mu_1 / \mu_2)^2}}. \quad (15)$$

Other Notable Angles of Incidence:

- At **normal incidence** ($\theta_i = 0$), $\Gamma_{\perp} = \Gamma_{\parallel}$.
- When $\theta_i = 90^\circ$, which is called **grazing incidence**, $|\Gamma_{\perp}| = |\Gamma_{\parallel}| = 1$.



Quick Review (Continued)

Suppose a circular beam of EM radiation shines on the boundary between two lossless media.

The **reflectivity** (or **reflectance**) R is the ratio of the reflected power over the incident power. For the **perpendicularly polarized** component, the reflectivity is

$$R_{\perp} = \frac{P_{\perp}^r}{P_{\perp}^i} = |\Gamma_{\perp}|^2. \quad (16)$$

In the case of **parallel polarization**, the reflectivity is

$$R_{\parallel} = \frac{P_{\parallel}^r}{P_{\parallel}^i} = |\Gamma_{\parallel}|^2. \quad (17)$$

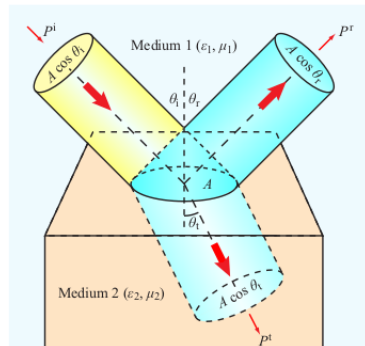


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

Quick Review (Continued)

The **transmissivity** (or **transmittance**) T is the ratio of the transmitted power over the incident power. For the **perpendicularly polarized** component, the transmissivity is

$$T_{\perp} = \frac{P_{\perp}^t}{P_{\perp}^i} = |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \quad (18)$$

In the case of **parallel polarization**, the transmissivity is

$$T_{\parallel} = \frac{P_{\parallel}^t}{P_{\parallel}^i} = |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \quad (19)$$

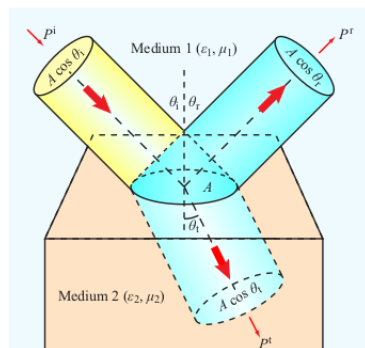


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

Quick Review (Continued)

Conservation of power implies

$$R_{\perp} + T_{\perp} = 1, \quad R_{\parallel} + T_{\parallel} = 1, \quad (20)$$

which are the same as

$$|\Gamma_{\perp}|^2 + |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1, \quad |\Gamma_{\parallel}|^2 + |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1. \quad (21)$$

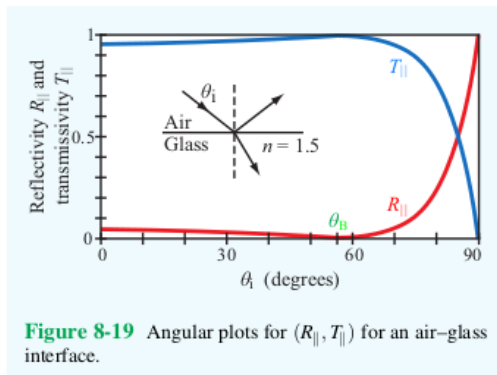


Figure 8-19 Angular plots for $(R_{\parallel}, T_{\parallel})$ for an air-glass interface.

Quick Review (Continued)

Table 8-2 Expressions for Γ , τ , R , and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2 . Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$
Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.			

Waveguides

Transmission lines can be divided into two categories. The **first category** contains lines that **support transverse electromagnetic (TEM) modes**, such as **coaxial cables, two-wire lines, and parallel-plate lines**.

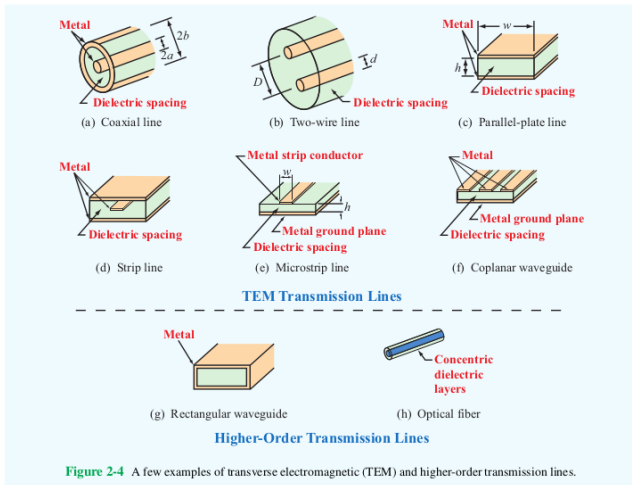


Figure 2-4 A few examples of transverse electromagnetic (TEM) and higher-order transmission lines.

Waveguides (Continued)

The **second category**, called **higher-order transmission lines**, consists of lines where one of $\mathbf{E}$ or $\mathbf{H}$ is orthogonal to the direction of propagation $\hat{\mathbf{k}}$ but not both. These include

- optical fibers
- metal waveguides

Both of these examples use **reflection** to guide waves in the desired direction. However, **optical fibers use total internal reflection** at the boundary between dielectric materials whereas **metal waveguides use the reflection at the boundary between a dielectric medium and a conductor**.

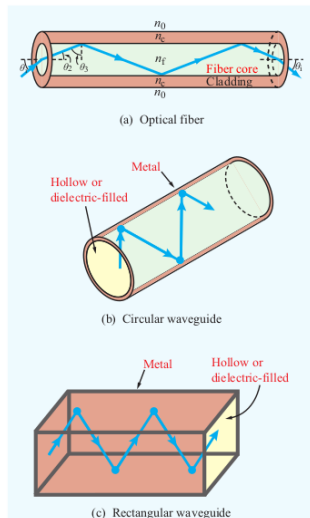


Figure 8-20 Wave travel by successive reflections in (a) an optical fiber, (b) a circular metal waveguide, and (c) a rectangular metal waveguide.

Waveguides (Continued)

In all **higher-order** transmission lines, **$\mathbf{E}$** or **$\mathbf{H}$** has a component in the **$\hat{\mathbf{k}}$** direction.

A **mode** is

- **transverse electric (TE)** if **$\mathbf{E}$** is orthogonal to **$\hat{\mathbf{k}}$** but not **$\mathbf{H}$** ,
- **transverse magnetic (TM)** if **$\mathbf{H}$** is orthogonal to **$\hat{\mathbf{k}}$** but not **$\mathbf{E}$** .

Signals can be **transferred between coaxial cables and metal waveguides** as shown in the figure. **Coaxial cables** are appropriate for transmitting frequencies **below 30 GHz** but **metal waveguides** are a good alternative for **high-power RF frequencies from 5 to 100 GHz**.

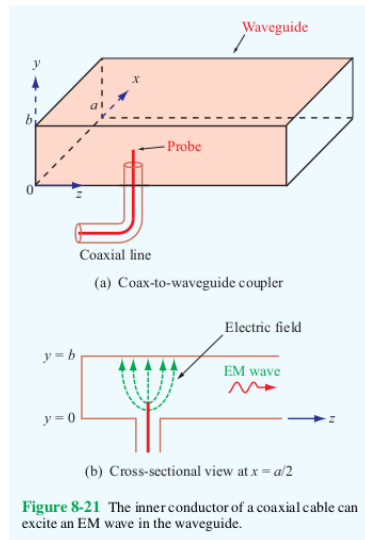


Figure 8-21 The inner conductor of a coaxial cable can excite an EM wave in the waveguide.

E and H in Rectangular Waveguides

The goal is to find expressions for **E** and **H** for the TE and TM modes in a rectangular waveguide. For this purpose, the coordinate system will be the one shown below.

In general,

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}} \quad (22)$$

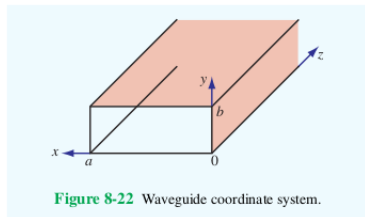
$$\tilde{\mathbf{H}} = \tilde{H}_x \hat{\mathbf{x}} + \tilde{H}_y \hat{\mathbf{y}} + \tilde{H}_z \hat{\mathbf{z}} \quad (23)$$

For a **wave traveling in the +z direction**, the components of the electric field phasors have the form

$$\tilde{E}_x(x, y, z) = \tilde{e}_x(x, y)e^{-j\beta z}, \quad (24)$$

$$\tilde{E}_y(x, y, z) = \tilde{e}_y(x, y)e^{-j\beta z}, \quad (25)$$

$$\tilde{E}_z(x, y, z) = \tilde{e}_z(x, y)e^{-j\beta z}, \quad (26)$$



E and H in Rectangular Waveguides (Continued)

where β is the **phase constant**.

The components of the magnetic field phasors have the same form. So,

$$\tilde{\mathbf{E}} = (\tilde{e}_x \hat{\mathbf{x}} + \tilde{e}_y \hat{\mathbf{y}} + \tilde{e}_z \hat{\mathbf{z}}) e^{-j\beta z} \quad (27)$$

$$\tilde{\mathbf{H}} = (\tilde{h}_x \hat{\mathbf{x}} + \tilde{h}_y \hat{\mathbf{y}} + \tilde{h}_z \hat{\mathbf{z}}) e^{-j\beta z}. \quad (28)$$

The interior of a metal waveguide is a **lossless, source-free medium**. In this case, the phasor form of Maxwell's equations is

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \quad (29)$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}}. \quad (30)$$

These equations can be written to express the x and y components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ in terms of their z components:

$$\tilde{E}_x = -\frac{j}{k_c^2} \left(\beta \frac{\partial \tilde{E}_z}{\partial x} + \omega\mu \frac{\partial \tilde{H}_z}{\partial y} \right), \quad \tilde{E}_y = \frac{j}{k_c^2} \left(-\beta \frac{\partial \tilde{E}_z}{\partial y} + \omega\mu \frac{\partial \tilde{H}_z}{\partial x} \right) \quad (31)$$

E and H in Rectangular Waveguides (Continued)

and

$$\tilde{H}_x = \frac{j}{k_c^2} \left(\omega\epsilon \frac{\partial \tilde{E}_z}{\partial y} - \beta \frac{\partial \tilde{H}_z}{\partial x} \right), \quad \tilde{H}_y = -\frac{j}{k_c^2} \left(\omega\epsilon \frac{\partial \tilde{E}_z}{\partial x} + \beta \frac{\partial \tilde{H}_z}{\partial y} \right), \quad (32)$$

where the **cutoff wavenumber** is

$$k_c^2 = k^2 - \beta^2 = \omega^2 \mu \epsilon - \beta^2 \quad (33)$$

and the **unbounded-medium wavenumber** is

$$k = \omega \sqrt{\mu \epsilon}. \quad (34)$$

The above equations mean that the z components $\tilde{E}_z$ and $\tilde{H}_z$ entirely determine the EM field phasors $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$.

TM Modes in a Rectangular Waveguide

In this case, $\tilde{\mathbf{H}}$ is orthogonal to the direction of propagation $\hat{\mathbf{z}}$, which means that $\tilde{H}_z = 0$ and the z **component of the electric field phasor $\tilde{E}_z$ entirely determines** the EM field phasors $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$.

The electric field phasor in an **unbounded, lossless medium** satisfies the **homogeneous wave equation**

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0. \quad (35)$$

This means that each component of $\tilde{\mathbf{E}}$ must satisfy this equation and, in particular,

$$\nabla^2 \tilde{E}_z + k^2 \tilde{E}_z = 0. \quad (36)$$

Here, $\tilde{E}_z(x, y, z) = \tilde{e}_z(x, y)e^{-j\beta z}$ and the above equation is equivalent to

$$\frac{\partial^2 \tilde{e}_z}{\partial x^2} + \frac{\partial^2 \tilde{e}_z}{\partial y^2} + k_c^2 \tilde{e}_z = 0. \quad (37)$$

This equation can be solved by assuming that

$$\tilde{e}_z(x, y) = X(x)Y(y). \quad (38)$$

TM Modes in a Rectangular Waveguide (Continued)

Then

$$\begin{aligned} 0 &= \frac{\partial^2}{\partial x^2} X(x) Y(y) + \frac{\partial^2}{\partial y^2} X(x) Y(y) + k_c^2 X(x) Y(y) \\ &= Y(y) \frac{d^2 X}{dx^2} + X(x) \frac{d^2 Y}{dy^2} + k_c^2 X(x) Y(y). \end{aligned} \quad (39)$$

Dividing by $X(x)Y(y)$, this implies

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + k_c^2 = 0. \quad (40)$$

Since k_c is a constant, the first term only depends on x , and the second term only depends on y ,

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2 \quad \text{and} \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2, \quad (41)$$

where k_x and k_y are constants with

$$k_x^2 + k_y^2 = k_c^2. \quad (42)$$

TM Modes in a Rectangular Waveguide (Continued)

Therefore,

$$\frac{d^2 X}{dx^2} + k_x^2 X = 0 \quad \text{and} \quad \frac{d^2 Y}{dy^2} + k_y^2 Y = 0. \quad (43)$$

These are second-order ODEs for $X(x)$ and $Y(y)$ with the solutions

$$X(x) = A \cos k_x x + B \sin k_x x \quad (44)$$

$$Y(y) = C \cos k_y y + D \sin k_y y, \quad (45)$$

where A , B , C , and D are constants, so that

$$\tilde{e}_z(x, y) = (A \cos k_x x + B \sin k_x x) (C \cos k_y y + D \sin k_y y). \quad (46)$$

The electric field decreases exponentially in the walls of the waveguide, which can be idealized by requiring $\tilde{e}_z = 0$ at the walls. This means imposing the **boundary conditions**

$$X(0) = X(a) = 0 \quad \text{and} \quad Y(0) = Y(b) = 0. \quad (47)$$

TM Modes in a Rectangular Waveguide (Continued)

The boundary conditions at $x = 0$ and $y = 0$ imply $A = C = 0$ and the boundary conditions at $x = a$ and $y = b$ imply

$$k_x = \frac{m\pi}{a}, \quad m = 1, 2, \dots \quad (48)$$

$$k_y = \frac{n\pi}{b}, \quad n = 1, 2, \dots \quad (49)$$

so that

$$\tilde{E}_z(x, y, z) = \tilde{e}_z(x, y)e^{-j\beta z} = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (50)$$

where E_0 is the constant amplitude of the electric field.

The **other components of the electric field phasor** and the **components of the magnetic field phasor** are found using the **expressions found earlier** with $\tilde{H}_z = 0$ and $\tilde{E}_z$ above. If either $m = 0$ or $n = 0$, $\tilde{E}_z = \tilde{H}_z = 0$ and all of the EM components are 0.

TM Modes in a Rectangular Waveguide (Continued)

Altogether, the mn^{th} **transverse magnetic mode** $\mathbf{TM}_{mn}$ has components

$$\tilde{E}_x(x, y, z) = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (51)$$

$$\tilde{E}_y(x, y, z) = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (52)$$

$$\tilde{E}_z(x, y, z) = E_0 \sin \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (53)$$

$$\tilde{H}_x(x, y, z) = \frac{j\omega\epsilon}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (54)$$

$$\tilde{H}_y(x, y, z) = -\frac{j\omega\epsilon}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (55)$$

$$\tilde{H}_z(x, y, z) = 0, \quad (56)$$

where the phase constant is

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2 \mu \epsilon - k_x^2 - k_y^2} = \sqrt{\omega^2 \mu \epsilon - (m\pi/a)^2 - (n\pi/b)^2}. \quad (57)$$

TM Modes in a Rectangular Waveguide (Continued)

The adjacent figure shows the fields of the mode TM_{11} .

When $k > k_c$, $\beta = \sqrt{k^2 - k_c^2}$ is real and the factor $e^{-j\beta z}$ corresponds to a wave propagating in the $+z$ direction. However, when $k < k_c$, β is imaginary and the exponent is real. In this case, the EM waves are **evanescent** and are quickly attenuated as z increases.

For each mode, i.e. each pair of values m and n , the **cutoff frequency**, where $\beta = 0$,

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (58)$$

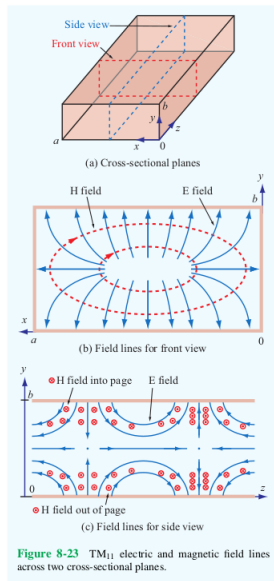


Figure 8-23 TM_{11} electric and magnetic field lines across two cross-sectional planes.

TM Modes in a Rectangular Waveguide (Continued)

is the **lower bound** on the frequencies of waves for that mode that will propagate through the waveguide. Here,

$$u_{p0} = 1/\sqrt{\mu\epsilon} \quad (59)$$

is the **speed of propagation in an unbounded medium**.

The **dominant mode** is the mode with the **lowest cutoff frequency**. This is TM_{11} for the TM modes and TE_{10} for the TE modes.

An **alternative expression** for β is

$$\beta = \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2} \quad (60)$$

and the **phase velocity** or **speed of propagation** of a TM or TE wave in the waveguide is

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad (61)$$

TM Modes in a Rectangular Waveguide (Continued)

Notice that the **phase velocity depends on the frequency**, even for waves with the same mode.

The **wave impedance** in the waveguide is

$$Z_{\text{TM}} = \frac{\tilde{E}_x}{\tilde{H}_y} = -\frac{\tilde{E}_y}{\tilde{H}_x} = \frac{\beta\eta}{k} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}, \quad (62)$$

where $\eta = \sqrt{\mu/\epsilon}$ is the intrinsic impedance of the dielectric medium.

TE Modes in a Rectangular Waveguide

The **transverse electric modes** can be found in a similar way but with $\tilde{E}_z = 0$. In this case, the mn^{th} **transverse electric mode** $\mathbf{TE}_{mn}$ has components

$$\tilde{E}_x(x, y, z) = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (63)$$

$$\tilde{E}_y(x, y, z) = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (64)$$

$$\tilde{E}_z(x, y, z) = 0 \quad (65)$$

$$\tilde{H}_x(x, y, z) = \frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (66)$$

$$\tilde{H}_y(x, y, z) = \frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) e^{-j\beta z} \quad (67)$$

$$\tilde{H}_z(x, y, z) = H_0 \cos \left(\frac{m\pi x}{a} \right) \cos \left(\frac{n\pi y}{b} \right) e^{-j\beta z}, \quad (68)$$

where β , f_{mn} , and u_p are given by the **same expressions** as before.

TE Modes in a Rectangular Waveguide (Continued)

In this case, m or n can be 0. By **convention**, a is defined to be the **longer length**, which means that TE_{10} is the **dominant mode**.

The **wave impedance** for TE modes is

$$Z_{TE} = \frac{\tilde{E}_x}{\tilde{H}_y} = -\frac{\tilde{E}_y}{\tilde{H}_x} = \frac{\eta}{\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}}. \quad (69)$$

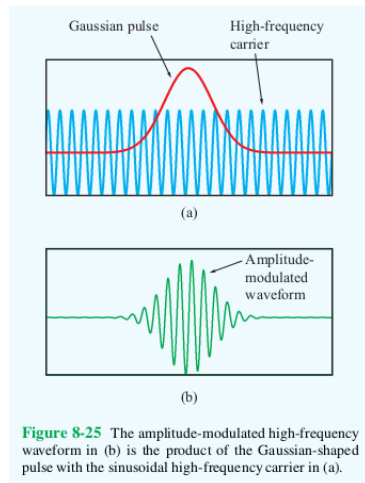
Summary of TM, TE Modes in a Rectangular Waveguide

Table 8-3 Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$, filled with a dielectric material with constitutive parameters ϵ and μ . The TEM case, shown for reference, pertains to plane-wave propagation in an unbounded medium.

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/Z_{TE}$ $\tilde{H}_y = \tilde{E}_x/Z_{TE}$ $\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{H}_x = -\tilde{E}_y/Z_{TM}$ $\tilde{H}_y = \tilde{E}_x/Z_{TM}$ $\tilde{H}_z = 0$ $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	$\tilde{E}_x = E_{x0} e^{-j\beta z}$ $\tilde{E}_y = E_{y0} e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/\eta$ $\tilde{H}_y = \tilde{E}_x/\eta$ $\tilde{H}_z = 0$ $\eta = \sqrt{\mu/\epsilon}$
<i>Properties Common to TE and TM Modes</i> $f_c = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ $\beta = k \sqrt{1 - (f_c/f)^2}$ $u_p = \frac{\omega}{\beta} = u_{p0} / \sqrt{1 - (f_c/f)^2}$		$f_c = \text{not applicable}$ $k = \omega \sqrt{\mu\epsilon}$ $u_{p0} = 1/\sqrt{\mu\epsilon}$

Propagation Velocities

Information is **encoded** into EM waves by modifying their **amplitude**, **frequency**, or **phase**. A modified wave has an **envelope**, which is the shape encapsulating the wave, and consists of the **frequency components** associated with its Fourier representation.



Propagation Velocities

In a **lossless, source-free, unbounded medium**,

- all frequency components **travel at the same speed**
- and the modified wave will travel indefinitely **without the envelope changing shape**.

But, in a **medium where the speed of propagation depends on frequency**,

- each frequency component will **travel at a different speed**
- and the **envelope will change shape over time**,

degrading the encoded information.

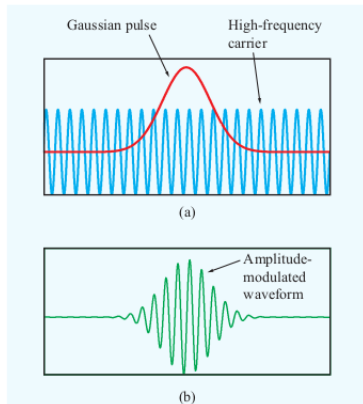


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Propagation Velocities (Continued)

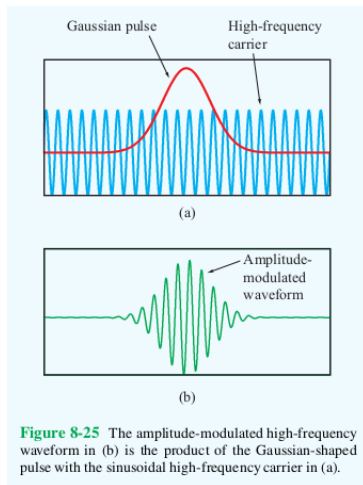
This phenomenon is **important in rectangular waveguides** since the **phase velocity**

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \quad (70)$$

has a **strong dependence on frequency**.

The **rate of information transmission** is characterized by the **speed** that the envelope or, equivalently, the **group of frequency components** constituting the wave, move in the medium. This speed is the **group velocity**

$$u_g = \frac{1}{d\beta/d\omega} = \frac{d\omega}{d\beta}. \quad (71)$$



Propagation Velocities (Continued)

For **TM or TE mode mn in a rectangular waveguide**,

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}. \quad (72)$$

Notice that, in this case,

$$u_p u_g = u_{p0}^2. \quad (73)$$

For the **waves that propagate through the waveguide**, $f > f_{mn}$. So,

- $u_p > u_{p0}$ and $u_g < u_{p0}$
- $\lim_{f \rightarrow \infty} u_p = \lim_{f \rightarrow \infty} u_g = u_{p0}$

Note that u_p can be $> c$ but u_g is always $\leq c$. Since energy and information travel at u_g , this is consistent with relativity.

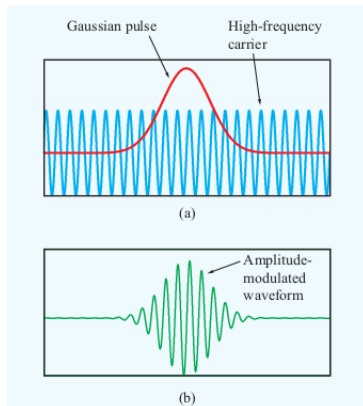


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Propagation Velocities (Continued)

An ω - β **diagram** is a plot of the relationship between ω and β . It provides useful insight into **how TM or TE modes propagate** in a waveguide compared to **how they would propagate in an unbounded medium**. Recall that

The ω - β curve for the mn mode comes from the formula

$$\begin{aligned}\beta &= \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2} \\ &= \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{2\pi f_{mn}}{\omega}\right)^2} \\ &= \frac{1}{u_{p0}} \sqrt{\omega^2 - (2\pi f_{mn})^2} \quad (74)\end{aligned}$$

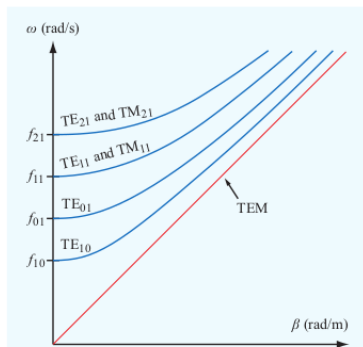


Figure 8-26 ω - β diagram for TE and TM modes in a hollow rectangular waveguide. The straight line pertains to propagation in an unbounded medium or on a TEM transmission line.

Propagation Velocities (Continued)

At a point on a curve in this diagram,

- $u_p = \omega/\beta$,
- $u_g = d\omega/d\beta$ is the slope of the curve.

A curve for mode mn has slope 0 and $\omega = 2\pi f_{mn}$ at $\beta = 0$. So, at this point, $u_g = 0$ and $u_p \rightarrow \infty$ as $\beta \rightarrow 0^+$.

The **TEM line** shows ω vs β when the medium is **unbounded**. On this line, $u_g = u_p = u_{p0}$ and the curves for all modes approach the TEM line as $f \rightarrow \infty$.

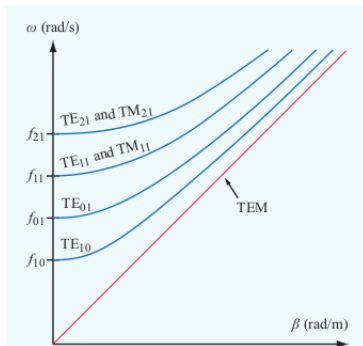
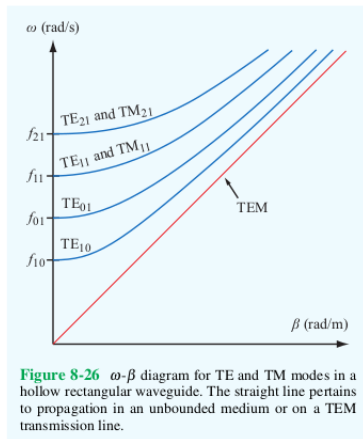


Figure 8-26 ω - β diagram for TE and TM modes in a hollow rectangular waveguide. The straight line pertains to propagation in an unbounded medium or on a TEM transmission line.

Propagation Velocities (Continued)

The above results pertain to **straight-line travel through the medium** in a waveguide and don't consider the consequences of reflection. Determining the **overall propagation velocity** requires accounting for the **zigzagging** of the path that a wave follows through the waveguide.



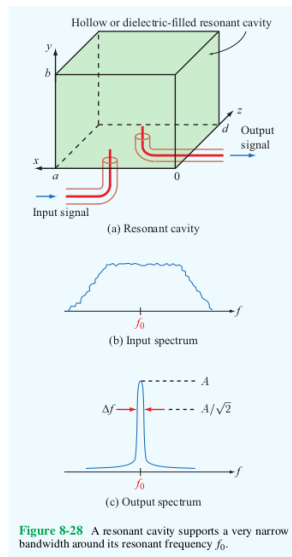
Cavity Resonators

A **cavity** is essentially a waveguide where the open ends have been replaced with metal walls. A cavity can be used as a microwave oscillator, an amplifier, or a bandpass filter.

Consider the $a \times b \times d$ rectangular cavity shown in the adjacent figure with the coordinate system shown. As discussed in the book, the boundary conditions in this case imply that

$$\beta = \frac{p\pi}{d}, \quad p = 1, 2, \dots \quad (75)$$

So, the modes in the cavity are specified by the three integers (m, n, p) .



Cavity Resonators

Each mode can only persist at the **resonant frequency**

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2} \quad (76)$$

For the TM modes, m and n start at 1 and p starts at 0. For the TE modes, m and n start at 0 and p starts at 1.

Although, in the ideal case, only the wave with frequency f_{mnp} persists in the cavity, a physical cavity will sustain waves with a small range of frequencies centered at f_{mnp} .

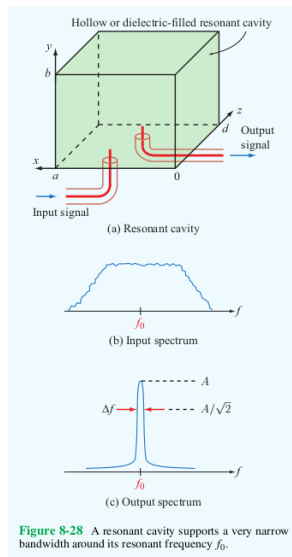


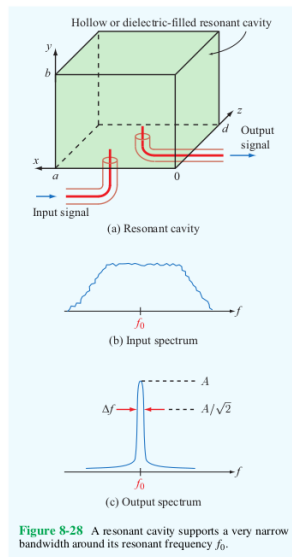
Figure 8-28 A resonant cavity supports a very narrow bandwidth around its resonant frequency f_0 .

Cavity Resonators

A measure of this is the **quality factor**

$$Q \approx \frac{f_{mnp}}{\Delta f}, \quad (77)$$

where Δf is measured at the point on the peak that is $f_{mnp}/\sqrt{2}$. Here, $\frac{\Delta f}{f_{mnp}}$ is the **normalized bandwidth** of the cavity.



Homework

Read Chapter 8.

Homework 3 is posted and due Tuesday, May 26th,
at the beginning of class.